Day 5 — Identifying & Removing Delivery Bottlenecks

> **Activity packet** for participant triads. Today's job: take the top queues from Day 4, design **bottleneck-removal experiments** with hypothesis / test / success criterion, integrate the full DP, peer-review, and ship.

Where we are in the week

The VSM exists (Day 4). Today turns its top queues into **experiments**: specific tests with specific success criteria, run on a real cadence. Then the full DP integrates and ships as the fifth and final sibling artifact.

By 16:00, every triad has the **complete sibling artifact set**: PRD / TCD / TMD / SEP / DP.

Inputs

- DP Sections 1–4
- The top 3 queues from Day 4
- The full sibling set (PRD / TCD / TMD / SEP)

The Theory of Constraints — one principle

Eli Goldratt's central claim: **a system's throughput is constrained by exactly one bottleneck at a time**. Not many. One. If you optimize anything other than the current bottleneck, you don't speed up the system — you just add inventory in front of the constraint.

For software delivery: if your slowest step is "code review takes 2 days," speeding up coding doesn't help — it just creates more code waiting for review. The TPM job is to **find the current bottleneck** and remove it; **then** find the new bottleneck.

This is also why "we're working on 5 improvements simultaneously" rarely works. Pick the constraint. Move it. Then pick the next one.

Bottleneck experiment — the structure

For each top queue, design an experiment:

```
### Experiment N: <short name>
**Bottleneck addressed:** <queue from VSM>

**Hypothesis:** We believe that <doing X> will <result>
because <reasoning>.

**Test:** <what we'll do, for how long, on what scope>

**Success criterion:** <measurable outcome that decides
adopt/reject>
- Baseline: <current measurement>
- Target: <improved measurement>
- Window: <duration>

**What could go wrong:**
<2–3 risks of running this experiment>

**Who runs it:** <named person>
**By when:** <date>

**Decision after:**
- If success criterion is met: <adopt / scale / institutionalize>
- If not met: <abandon / iterate / escalate>
```

The experiment **must** have a success criterion that's measurable. "We'll see if it feels better" is not an experiment — it's a vibe.

Activity 1 — Design 3 Experiments

Format: Triad • **35 min** • Block 1

Purpose

Convert the top 3 queues from Day 4 into 3 specific experiments.

Triad protocol

1. **Re-read your top 3 queues** (5 min) from DP Section 4.
2. **For each, draft an experiment** (25 min) using the template above. Commit to:
    - A specific change (not "improve code review" — "auto-assign reviewers based on path")
    - A duration (typically 2–4 sprints)

    - A success criterion with baseline, target, and window
3. **Sanity-check** (5 min). For each: would your team actually try this in the next 60 days? If not, the experiment is too ambitious or too vague.

Worked example — FieldPulse

```
### Experiment 1: Auto-assign code reviewers

**Bottleneck:** Build → Code review queue (1.5 days average wait).

**Hypothesis:** We believe that auto-assigning code reviewers
based on file paths (using CODEOWNERS) will reduce time-to-first-
review from 1.5 days to ≤ 4 hours, because the current 1.5-day
wait is mostly "who's going to review this" decision time.

**Test:** Configure CODEOWNERS for the reconcile module; run
for 4 sprints (8 weeks); compare time-to-first-review and time-
to-merge against the 8 sprints prior.

**Success criterion:**
- Baseline: time-to-first-review p50 = 1.5 days, p90 = 3 days
- Target: p50 ≤ 4 hours, p90 ≤ 1 day, over the 4 sprints
- Window: 8 weeks; min sample 30 PRs

**What could go wrong:**
- Reviewers feel burdened by constant assignment → measure complaints /
        opt-outs
- Auto-assigned reviewer is on PTO → fall-back to next-in-list
- One reviewer becomes overloaded → measure per-reviewer load

**Who runs it:** <eng lead> with <TPM> support
**By when:** Pilot starts next sprint (2026-05-13)

**Decision after:**
- Met: institutionalize CODEOWNERS for the whole codebase
- Not met: examine why; possibly try paired-review during dev
```

Activity 2 — DP Integration Pass

Format: Triad • **40 min** • Block 2

Purpose

Read the full DP top to bottom. Fix incoherences. Lock the version that goes into Friday's review.

The integration checklist

Check	Pass criterion
Section 1 outcomes ↔ Section 2 backlog	Every output in Section 2 maps to an outcome in Section 1
Section 2 backlog ↔ Section 3 tracking	Every leading indicator in Section 3 corresponds to backlog items
Section 3 tracking ↔ Section 4 VSM	Cycle-time leading indicators are consistent with VSM measurements
Section 4 queues ↔ Section 5 experiments	Every top queue has at least one experiment
References to PRD/TCD/TMD/SEP	DP cites prior artifacts by section
No fortune-cookie prose	Specific to this feature

Triad protocol

1. **Solo read-through** (15 min). Each member reads Sections 1–5; marks issues in margins.
2. **Pool issues** (10 min). De-dupe.
3. **Fix in priority order** (15 min). Coherence first, then prose.

Activity 3 — Cross-Review with Friday Rubric

Format: Triad-pair • **40 min** • Block 3

Purpose

Cross-review the full DP with another triad using the Friday rubric.

The Friday rubric (DP)

Dimension	Weight	Exemplary
Outcome clarity	20%	Outcomes user-visible; tied to NS / Tier Sheet
ADO discipline	15%	Hierarchy, fields, tags, iterations populated; queries work
Output ↔ outcome pairing	20%	Each output has outcome AND leading indicator
Value stream realism	20%	Lead times measured (or honestly estimated); queues named
Bottleneck experiments	15%	Each has hypothesis, test, success criterion
Integration with prior artifacts	10%	DP references PRD/TCD/TMD/SEP appropriately

Score 0–4 per dimension. Apply weights for total /4.0. Same convention as prior weeks.

Cross-review protocol (40 min)

1. **Pair triads** (5 min). Same pair as prior weeks if possible.
2. **20 min cross-read** with the rubric.
3. **10 min author-triad responds** to feedback.
4. **5 min final score** captured.

Activity 4 — Sign-Off + Bridge to Week 8

Format: Triad • **45 min** + Wrap • Block 4

Purpose

Sign off the DP. Bridge to Week 8's capstone work.

Sign-off

The triad declares the DP **Status: Approved** (or "Approved with gaps") — same convention as prior weeks. The DP joins PRD / TCD / TMD / SEP as the fifth sibling artifact.

Bridge to Week 8 — what the capstone needs

Week 8 is the capstone. Each triad picks a capstone subject — **FieldPulse or a real project of their own** (work, school, or hobby). They produce an integrated artifact set in shorter form, anchored on the Week-8 topic: **AI Spec Development**.

What carries over to Week 8:

- The artifact templates (PRD/TCD/TMD/SEP/DP) — used in shorter form
- The AI provenance discipline
- The negotiation / outcome / tracking muscles
- The cumulative AI-validation log

What's new in Week 8:

- **AI Spec Development** — using AI to draft technical specs that integrate with the artifact set
- A **capstone subject** (FieldPulse or a project of your own)
- **Final artifact presentations** Friday

Triad protocol — Week 8 readiness check (30 min)

1. **Pick a capstone candidate** (15 min). Brainstorm 3–5 candidate subjects (FieldPulse or real projects of your own) for Week 8. Criteria:
    - Real product / feature you've encountered (work, school, hobby, public)
    - Substantive enough for a week of work
    - A clear problem you can state in one sentence
    - Permission to discuss publicly (no NDA conflicts)
2. **Pre-flight** (10 min). What inputs would you need? Who would you ask?
3. **Declare a candidate** (5 min). Triad's preferred Week-8 capstone. Bring it Monday.

Final wrap (last 15 min)

Each triad shares:

- The **bottleneck experiment** they're most excited to run
- The **single sharpest insight** the DP added beyond the rest of the artifact set
- Their **declared Week-8 capstone candidate**

End-of-week (Week 7) checkpoint

Each triad ships:

- [x] **Full DP** with Sections 1–5
- [x] Section 1 Outcome map
- [x] Section 2 Loaded ADO backlog (or paper equivalent) with field discipline
- [x] Section 3 Tracking plan (output / outcome / leading indicator triples + cadences)
- [x] Section 4 Value stream map with PT / LT / flow efficiency

- [x] Section 5 Bottleneck removal experiments (hypothesis / test / success criterion)
- [x] AI provenance log entries
- [x] Cross-reviewed + signed off
- [x] **Week 8 capstone candidate declared**

The DP joins the PRD / TCD / TMD / SEP. Together they are the **complete sibling artifact set** built across Weeks 3–7 — the input to the Week-8 capstone.